4. Exercise

Linear Models and Regression

Exercise Sheet 4

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Solve the following problems before the exercise class on December 4. Use `R` for all computations and all statistical tests.

Problem 1
The data set `Greenberg.csv`, which was uploaded to Stud.IP, provides for children the height, the information if they attended a private or a rural school and the age. One is interested in explaining the height. Use `R` for the following analyses:

a) Consider the linear model with the school as only independent variable and study whether there is a difference between the two schools. Formulate the null hypothesis and the alternative and conduct a test at level 0.05.

b) Take for the linear model the school and the age of the children into account and investigate if there is a difference between the two schools. Formulate the null hypothesis and the alternative and test at level 0.05.

Problem 2
The data set `PlantGrowth` in `R` compares the yields of plants (described by the weight) under three different conditions. In the following, we consider analysis of variance.

a) Check if there is a difference between the three groups. Formulate the null hypothesis and the alternative and use a test with level 0.05.

b) Decide which groups are different. Formulate the test problems and use the Tukey-Kramer test with level 0.05.

Problem 3
The data set `Salaries` from the `R` package `car` provides the salaries of professors in the US. Our goal is to study how the quantities `rank` (with levels `AsstProf`, `AssocProf` and `Prof`), `discipline` (with levels `A` and `B`), `yrs.since.phd` and `sex` (with levels `Female` and `Male`) affect the so-called nine-month salary `salary`.
Consider a linear model with the variables given above. For all statistical tests it is supposed that the assumptions of a Gaussian linear model are satisfied. Use the level 0.05 for all tests and do not correct for multiple testing.

a) Estimate the intercept and the coefficients of the corresponding linear model as well as the standard deviation σ of the errors.

b) Decide which of the independent variables are relevant for the salary. Provide the null hypotheses and the alternatives and use suitable tests.

Problem 1

Recall

$$Y = \gamma_0 + \gamma_1 x_1 + \gamma_2 x_2 + \gamma_3 \mathbb{1}_{\{z=2\}} + \gamma_4 \mathbb{1}_{\{z=3\}}$$

$\mathbb{1}$

z - factor - levels - indicator function

↳ leave out the first level

⇒ height ↓ dependent r.v.

school ↓ independent r.v. { private, public }

a) We assume

$$\text{height} = \gamma_0 + \gamma_1 \mathbb{1}_{\{\text{school} = \text{rural}\}} + \varepsilon$$

with $\varepsilon \sim \text{Normal}(0, \sigma^2)$

We test

$H_0: \gamma_1 = 0$ vs $\gamma_1 \neq 0$ ← always write in H_1 ↓ what we want to prove?

H_0 — p-value — α — H_1 reject

curvet p-value from R → 0.298.

Since p-value 0.298 is greater than 0.05, we cannot reject H_0 at level 0.05

b) We assume

$$\text{height} = \gamma_0 + \gamma_1 \mathbb{1}_{\{\text{school} = \text{rural}\}} + \gamma_2 \cdot \text{age} + \varepsilon$$

with $\varepsilon \sim \mathcal{N}ormal(0, \sigma^2)$

We test

$$H_0 : \gamma_1 = 0 \quad \text{vs} \quad H_1 : \gamma_1 \neq 0$$

p-value from R $\rightarrow$ 0.042

Since the p-value 0.042 is less than 0.05, we can reject H_0 at level 0.05

Problem 2

a) We test

$$H_0 : \mu_{crte} = \mu_{trt1} = \mu_{trt2}$$

$$\text{vs} \quad H_1 : \mu_i \neq \mu_j \text{ for some } i, j \in \{crte, trt1, trt2\}, \; i \neq j$$

Since the p value 0.0159 is less than 0.05, we can reject H_0 i.e., there is no difference between the 3 groups at level 0.05.

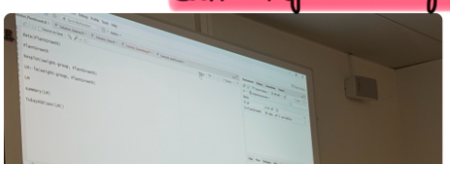
b) We test

$$H_0^{1,2} : \mu_{crt1} = \mu_{trt1} \quad \text{vs} \quad H_1^{1,2} : \mu_{ctrl} \neq \mu_{trt1}$$

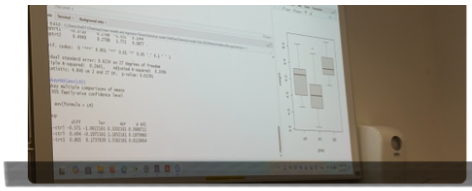
$$H_0^{1,3} : \mu_{crt1} = \mu_{trt1} \quad \text{vs} \quad H_1^{1,3} : \mu_{ctrl} \neq \mu_{trt2}$$

$$H_0^{2,3} : \mu_{trt1} = \mu_{trt2} \quad \text{vs} \quad H_1^{2,3} : \mu_{trt1} \neq \mu_{trt2}$$

Only the p-value of the 3rd problem is less than 0.05, we can reject $H_0^{2,3}$



$$\mu_{trt1} < \mu_{trt2}$$



↳ compare coefficients
since coefficient of trt2 > trt1

Problem 3

a) We consider the linear model

$$\text{Salary} = \gamma_0 + \gamma_1 \mathbb{1}\{\text{rank} = \text{Assoc. Prof}\} + \gamma_2 \mathbb{1}\{\text{rank} = \text{Prof}\} + \gamma_3 \mathbb{1}\{\text{discipline} = B\}$$
$$+ \gamma_4 \text{ yrs.since.phd} + \gamma_5 \mathbb{1}\{\text{sex} = \text{male}\} + \varepsilon.$$

with $\varepsilon \sim \text{Normal}(0, \sigma^2)$

b) We test

Anova

				p-value
$H_0^{(1)}$:	$\gamma_1 = \gamma_2 = 0$	vs	$H_1^{(1)}$: $\gamma_1 \neq 0$ & $\gamma_2 \neq 0$	2.2×10^{-6} ↳ reject
$H_0^{(2)}$:	$\gamma_3 = 0$	vs	$H_1^{(2)}$: $\gamma_3 \neq 0$	6.3×10^{-9} ↳ reject
$H_0^{(3)}$:	$\gamma_4 = 0$	vs	$H_1^{(3)}$: $\gamma_4 \neq 0$	0.63124 ↳ cannot reject
$H_0^{(4)}$:	$\gamma_5 = 0$	vs	$H_1^{(4)}$: $\gamma_5 \neq 0$.	0.26242 ↳ cannot reject

c) We test

$H_0 : \gamma_1 = 0$ vs $H_1 : \gamma_1 \neq 0$

Since the p-value is 0.00179
we can reject H_0 at level 0.05.

$\hat{\gamma}_1 = 13104.5 > 0$

↳ positive.

⇓

Salary higher
for associate professor.

#Problem 4

$$Y = \gamma_0 + \sum_{i=1}^{m} \gamma_i x_i + \sum_{j=1}^{l-1} \gamma_{m+j} \mathbb{1}_{\{t=j+1\}} + \varepsilon$$

$$\gamma_0, \gamma_1, \dots, \gamma_m, \gamma_{m+1}, \dots, \gamma_{m+l-1}$$

Testing

$$H_0: \gamma_{m+1} = \dots = \gamma_{m+l-1} = 0$$

$$\text{vs} \quad H_1: \gamma_j \neq 0 \text{ for some } j \in \{m+1, \dots, m+l-1\}$$

$\Leftrightarrow$

$$H_0: \gamma_j = 0 \text{ for all } j \in \{0, \dots, m+l-1\} \setminus J$$

against $H_1: \gamma_j \neq 0$ for some $j \in \{0, \dots, m+l-1\}$

where $J = \{0, \dots, m\}$

From corollary 3.14, a level α test for the problem is given by

$$\tilde{\varphi}(Y) = \begin{cases} \text{cannot reject} & \dfrac{(n-(m+l-1)-1)\left(\|\hat{\varepsilon}_J\|^2 - \|\hat{\varepsilon}\|^2\right)}{((m+l-1)+1-(m+1))\,\|\hat{\varepsilon}\|^2} < F_{(m+l-1)+1-(m+1),\, n-(m+l-1)-1,\, 1-\alpha} \\ \text{reject} & \text{Otherwise} \end{cases}$$

$$Y = \tilde{A}\hat{\gamma} + \hat{\varepsilon}_J$$

$$\hat{\gamma} = (\tilde{A}^T\tilde{A})^{-1}\tilde{A}^T Y$$

$$\hat{\varepsilon}_J = Y - \tilde{A}\hat{\gamma}$$
$$= Y - \tilde{A}(\tilde{A}^T\tilde{A})^{-1}\tilde{A}^T Y$$

$$\varphi(Y) = \begin{cases} \text{cannot reject} & \dfrac{(n-m-l)\left(\|Y-\tilde{A}(\tilde{A}^T\tilde{A})^{-1}\tilde{A}^T Y\|^2 - \|Y - A(A^TA)^{-1}A^TY\|^2\right)}{(l-1)\|Y - A(A^TA)^{-1}A^TY\|^2} < F_{l-1,\,n-m-l,\,1-\alpha} \\ \text{reject} & \text{otherwise.} \end{cases}$$

So φ is a level-α test